EE5203 L13 Worksheet

Defence rehearsal - Basri Erdogan

Expected time: 30-45 minutes **Policy:** This worksheet is your defence rehearsal. Answer for **your own project and your own contribution** — generic answers earn nothing here.

Do Part E with a teammate, out loud, before the session.

Part A - Your contribution

1. List the parts of the system **you** personally wrote or built. For each, name the resources it owns (pins, timers, peripherals).
2. For one of those parts, write the walkthrough you would give in 90 seconds: what it does, how, and where it sits in the system.
3. Which requirement is your part responsible for? State it in testable form and give the test.
4. What does your part cost in the timing budget? Show the arithmetic.

Part B - Mechanism

5. Name three registers your code configures. For each, give the value and say what the hardware does as a result.

Register	Value	What the hardware does

6. Pick one line of your configuration code. Write down precisely what would happen if it were deleted — the symptom, not the abstraction.
7. Your system uses at least three peripherals that raise flags. Name each flag, what sets it, and what clears it.
8. Where in your system is there an interrupt callback? State what it records and what `main()` does with that. If a callback does application work, explain why that is or is not acceptable here.

Part C - Decisions

9. Turn each fact into a design decision with its rejected alternative.

Fact	Decision (with the reason)
"We used interrupts for the button."	
"We send data over UART."	
"We poll the sensor every 10 ms."	

10. Write three design decisions from **your own** project in the same form: what you chose, what you rejected, and why.
11. Identify one decision you now think was wrong. What would you choose instead, and what would it have cost to change at the time?

Part D - Evidence and failure

12. For each requirement in your project, complete the row.

#	Requirement (testable)	Test procedure	Result observed
1			
2			
3			

13. Which fault will you inject during the demonstration? State the prediction you will make beforehand, and the evidence you will show afterwards.

14. Write your known-issues list. For each: the symptom, what you know about the cause, whether you attempted a fix, and the impact.
15. Your demonstration fails in the first minute. Write the four sentences you would say, in order.

Part E - Mock defence

Do this **out loud, with a teammate**, timed. Five minutes each.

16. Have them ask you these, in a random order, and answer without notes:

- Which parts did you write, and which resources do they own?
- Show me where you configure that peripheral. Why those values?
- What is in that register right now, and why?
- What would happen if I removed this line?
- How do you know this requirement is met? Show me the evidence.
- What broke during the term, and how did you find it?

17. Note the two questions you answered worst. Write proper answers now.
18. Have them ask one question you did **not** expect. Record it — that is the one worth preparing.

Part F - Reflection

19. What would you design differently, knowing what you know now? Three sentences, specific enough that another team could act on them.
20. What cost you the most time this term, and what would have prevented it? Three sentences.

Exit self-assessment

- ☐ I can name my own contribution and the resources it owns.
- ☐ I can explain three registers my code configures.
- ☐ I can state three design decisions with rejected alternatives.
- ☐ Every requirement I own has a test and an observed result.
- ☐ I have rehearsed the six defence questions out loud.
- ☐ Both reflection answers are written and specific.